

A Soft Introduction to CMA-ES Technique

Optimization Team

Seangleng Khe (KHM)

other names

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Instructor

Prof. Gaél Raoul

SEAMS School 2026: Applying Mathematics to Real-World Applications

University of Science Ho Chi Minh, Vietnam

Introduction

The objective of the study is to understand the progression

random search → evolution strategy → adaptive Gaussian search → CMA-ES

After: we shall see how important it is to adapt a full covariance matrix can make the algorithm nearly insensitive to rotations and anisotropic scalings of the search landscape.

Derivative-Free Optimization

Let $D \subset \mathbb{R}^d$ and $f : D \rightarrow \mathbb{R}$. We want to minimize the function f

$$\min_{x \in D} f(x) \quad (1)$$

If f is smooth (i.e., continuously differentiable) then to solve (1), we can use gradient descent methods, such as steepest descent method or Newton-like methods, which can generally generate the solution by updating at each iterate

$$x_{n+1} = x_n - \sigma \nabla f(x_n)$$

Challenges: f is not differentiable!

We are here to answer three questions:

1. What information does a gradient use that a population-based methods does not have directly?

Ans Gradient methods use the first-order derivative of the objective function. The gradient provides the descent direction in the search process to obtain the minimizer which the population-based methods do not access. Derivative-free methods evaluate the objective function values at the a set of candidate points and select the best that improves the function value. These methods however, help us to do optimization with nonsmooth functions which is practical in real-world applications.

2. Why can derivative-free methods still recover preferred directions of descent?

Ans Yes, DFO can recover the descent direction even without gradient information. To obtain so, DFO implicitly approximate the gradient-like information using the fitness values (function values) and they structure their search so that at least one trial direction is guaranteed to be descent direction.

3. What difficulties arise when d is large?

(μ, λ) -ES

Algorithm 1: $(1, \lambda)$ -ES Iteration: Isotropic Search

Input: A point $m_0 \in \mathbb{R}^d$, step size $\sigma > 0$, population size λ

```
1 for  $k \leftarrow 1$  to  $n$  do
2   for  $i \leftarrow 1$  to  $\lambda$  do
3     Sample  $Z_i \sim \mathcal{N}(0, I_d)$ 
4     Compute  $X_i \leftarrow m_k + \sigma Z_i$  //  $X_i \sim \mathcal{N}(m_0, \sigma^2 I_d)$ 
5   Compute fitness value  $f(X_1), \dots, f(X_\lambda)$ 
6    $i^* \leftarrow \arg \min_{i \in \{1, \dots, \lambda\}} f(X_i)$ 
7    $m_{k+1} \leftarrow X_{i^*}$ 
8 return  $m_n$ 
```

Algorithm 2: (μ, λ) -ES Iteration: Isotropic Search

Input: A point $m_0 \in \mathbb{R}^d$, step size $\sigma > 0$, population size λ , number of selected parent $\mu \leq \lambda$, weighted factor ω_i s.t. $\sum_{i=1}^{\mu} \omega_i = 1$.

1 **for** $k \leftarrow 1$ **to** n **do**

2 **for** $i \leftarrow 1$ **to** λ **do**

3 $Z_i \sim \mathcal{N}(0, I_d)$

4 $X_i = m_k + \sigma Z_i$

5 Compute $f(X_i)$ and sort offspring by fitness:

$$X_{1:\lambda}, X_{2:\lambda}, \dots, X_{\lambda:\lambda},$$

where $X_{1:\lambda}$ is the best, $X_{2:\lambda}$ is the second-best, and so on.

6 Choose μ best candidates and recombine

$$m_{n+1} = \sum_{i=1}^{\mu} \omega_i X_{i:\lambda}.$$

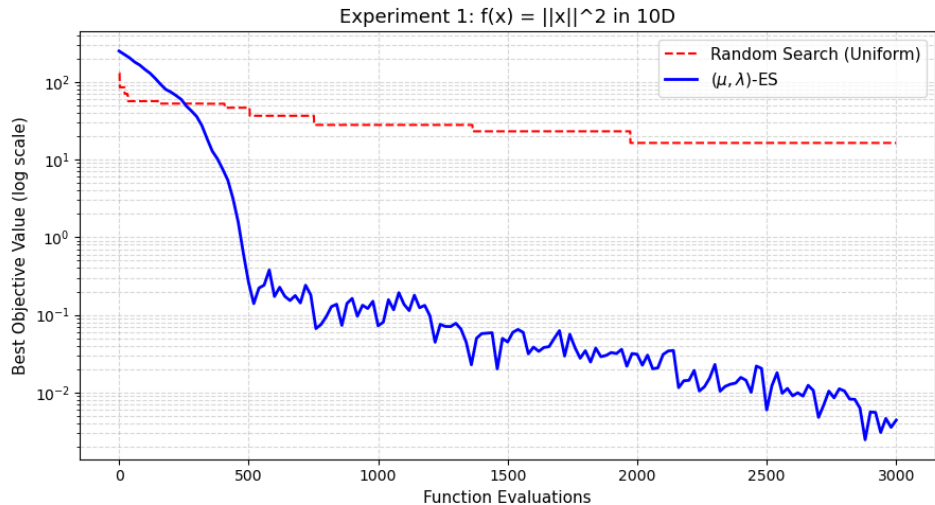


Figure 1: Comparison between random search with uniform distribution and (μ, λ) -ES.

III-Conditioned Quadratic

Anisotropic level set

- With covariance adaptation, we see that to obtain the solution it go through to the optimal solution.
- While the isotropic search cannot. It more or less is taking a detour to obtain the solution.
- A central observation is that isotropic mutation is inefficient when the successful search from a thin elongated ellipsoid.

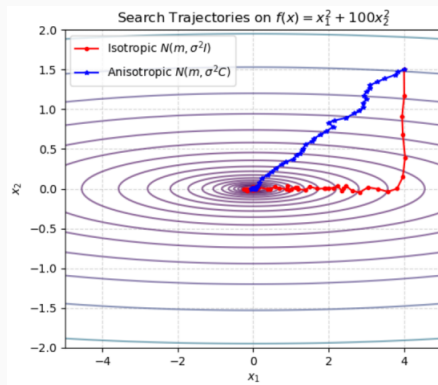


Figure 2: Function f has anisotropic level set. (Steep valey in x_2 and flat in x_1).

This leads us to **anisotropic search**.

In stead of sampling from

$$X \sim \mathcal{N}(m, \sigma^2 I),$$

we consider the offspring of the form

$$X \sim \mathcal{N}(m, \sigma^2 C)$$

where C is symmetric positive definite.

